

Breast Cancer Classification using Machine Learning

Task 4 – Metric Selection, Imbalance Handling and Final Model Decision

1. Objective

The objective of this project is to develop and evaluate machine learning classification models for predicting the class of breast cancer samples. Multiple classification algorithms are trained and compared using Accuracy, Precision, Recall, F1 Score and ROC-AUC. The project also investigates class imbalance and evaluates a class-weighted Logistic Regression model before selecting the final model for deployment.

2. Dataset

The project uses the Breast Cancer Wisconsin (Diagnostic) dataset available through Scikit-learn. The dataset contains 569 instances and 30 numerical predictive features. The features represent numerical measurements derived from characteristics of breast-cell nuclei. The target contains two classes: malignant and benign.

3. Classification Models

Three classification approaches were evaluated. Logistic Regression was used as the baseline model. A second Logistic Regression model was trained using `class_weight="balanced"` to handle differences in class frequency. A Decision Tree Classifier was also trained to provide a comparison with a tree-based model.

4. Metric Selection

Accuracy alone is not sufficient for evaluating a classification model, especially when the target classes may not be equally represented. Therefore, several complementary metrics were selected.

Accuracy: measures the proportion of all predictions that are correct. **Precision:** measures how many observations predicted as positive are actually positive. **Recall:** measures how many of the actual positive observations are correctly identified.

F1 Score: combines Precision and Recall into a single score using their harmonic mean. It is useful when both false positives and false negatives matter. **ROC-AUC:** evaluates how well the model separates the two classes over different probability thresholds.

5. Confusion Matrix

The confusion matrix provides a detailed view of classification errors. It contains True Positives, True Negatives, False Positives and False Negatives. Examining these values helps identify the type of errors a model makes instead of relying only on the overall accuracy.

6. ROC Curve and ROC-AUC

The ROC curve is created from predicted probabilities. It plots the True Positive Rate against the False Positive Rate at different classification thresholds. The Area Under the Curve (AUC) summarizes the model's ability to distinguish between the two classes. A value close to 1 indicates strong class separation.

7. Class Imbalance and Its Handling

Class imbalance occurs when the number of observations in one class differs from the number in another class. If imbalance is present, a model can sometimes obtain a high accuracy while performing poorly for the less-represented class.

To account for this possibility, the project evaluates performance using Precision, Recall and F1 Score in addition to Accuracy. A balanced Logistic Regression model is also trained using `class_weight="balanced"`. This makes the learning algorithm give greater importance to classes that occur less frequently.

8. Model Performance Comparison

Model	Accuracy	Precision	Recall	F1	AUC
Logistic Regression	95.61%	94.59%	98.59%	96.55%	99.77%
Balanced Logistic Regression	97.37%	97.22%	98.59%	97.90%	99.77%
Decision Tree	94.74%	95.77%	95.77%	95.77%	94.40%

Balanced Logistic Regression produced the strongest overall performance. It achieved 97.37% Accuracy, 97.22% Precision, 98.59% Recall, 97.90% F1 Score and 0.9977 ROC-AUC.

9. Final Model Decision

The final model was selected by considering the complete set of evaluation metrics rather than accuracy alone. Balanced Logistic Regression achieved the highest Accuracy, Precision and F1 Score among the tested models while maintaining a high Recall of 98.59%. Its ROC-AUC of approximately 0.9977 also indicates excellent class separation on the evaluated test data.

Therefore, **Balanced Logistic Regression** was selected as the final model for the project.

10. Final Model Verification

The selected Balanced Logistic Regression model was saved using Joblib as `best_classification_model.pkl`. The saved model was loaded again and tested on the test dataset. It successfully generated both class predictions and probability predictions.

The loaded model reproduced the same ROC-AUC value of **0.9977071732721913**. This verifies that the model saved for deployment is consistent with the model evaluated during the experimentation stage.

11. Streamlit Application and Deployment

A Streamlit application was developed using the saved model. The application provides an interactive interface where the required numerical feature values can be supplied and the trained model generates a classification prediction together with probability information.

The project source code and supporting files are maintained in a GitHub repository and the Streamlit application has been deployed using Streamlit Community Cloud.

12. Conclusion

This project demonstrates a complete binary classification workflow. It includes dataset preparation, model training, evaluation using complementary metrics, examination of class imbalance, class-weighted Logistic Regression, Decision Tree comparison, model selection, model persistence and Streamlit deployment.

Balanced Logistic Regression was selected as the final model because it provided the best overall combination of the evaluated metrics. Its final test performance was 97.37% Accuracy, 97.22% Precision, 98.59% Recall, 97.90% F1 Score and 99.77% ROC-AUC.

Educational Disclaimer: This project is an academic machine learning demonstration and is not a medical diagnostic system. Predictions from this application should not be used for medical decisions.